DT_DMS_Architecture

DT_DMS
$DT = \langle D, M, S, \mu \rangle$

D 0..*

M 0..*

μ 1

S 0..*

Mu

«interface»
ObserverDataManager

«interface»
ObserverData

«interface»
ObserverModel

DT_Mu
$\mu : \Omega_D \times M \to M$

Automaton

«abstract»
DT_Automaton
$\langle \Lambda, X, Y, \Sigma, \Phi, \Delta, \Delta^* \rangle$

observes

notifies 0..*

updates: μ(ω, m) = m′

Data

DT_DataCharacterization
unit, signal, sensor, ...

1

1

DT_Data
$D = D_{hist} \cup D_{rt}$

DT_DataSegment
$\omega : T_U \rightharpoonup D$

Models

DT_Model_Predictive (M_{pred})

«abstract»
DT_Model
$m \in M$

DT_Model_Descriptive (M_{desc})

DT_Model_Prescriptive (M_{presc})

Services

coupled model 0..1

DT_Service_Control (S_{cont})

DT_Service_Optimization (S_{opti})

«abstract»
DT_Service
$s : M \times \Omega_D \to Y_s$

DT_Service_Diagnosis (S_{diag})

DT_Service_Prediction (S_{pred})

DT_Service_Monitoring (S_{moni})